Reduction of order

Remember from the previous lesson that we identified three possible forms for the general solution of a second order homogeneous linear differential equation.

Roots	General solution
Distinct real roots	$y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$
Equal real roots	$y(x) = c_1 e^{r_1 x} + c_2 x e^{r_1 x}$
Complex conjugate roots	$y(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$

We learned to use the characteristic equation associated with the second order equation to solve for the roots r_1 and r_2. But it's important to say that, while these values for r are the roots of the characteristic equation, they aren't by themselves solutions to the differential equation.

Solutions of the differential equation

Instead, we need to identify the solutions to the differential equation from the equation of the general solution, given in the table above. The solutions to the differential equation will be the functions we see multiplied by c_1 and c_2.

For instance, given a general solution with distinct real roots, $y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$, the solutions to the differential equation are

$$y_1 = e^{r_1 x}$$

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$$y_2 = e^{r_2 x}$$

Alternatively, given a general solution with complex conjugate roots,

$$y(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

$$y(x) = c_1 e^{\alpha x} \cos(\beta x) + c_2 e^{\alpha x} \sin(\beta x)$$

the solutions to the differential equation are

$$y_1 = e^{\alpha x} \cos(\beta x)$$

$$y_2 = e^{\alpha x} \sin(\beta x)$$

Notice how all three forms of the general solution given in the table above can actually be written in the same form, $y(x) = c_1 y_1 + c_2 y_2$.

	Solutions of the equation
$y(x) = c_1 y_1 + c_2 y_2$	
$y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$	$\{y_1, y_2\} = \{e^{r_1 x}, e^{r_2 x}\}$
$y(x) = c_1 e^{r_1 x} + c_2 x e^{r_1 x}$	$\{y_1, y_2\} = \{e^{r_1 x}, x e^{r_2 x}\}$
$y(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$	$\{y_1, y_2\} = \{e^{\alpha x} \cos(\beta x), e^{\alpha x} \sin(\beta x)\}$

Now that we've identified this set of two solutions for each form of a second order linear homogeneous equation, let's explain what it actually means for y_1 and y_2 to be solutions of the original equation.

Put plainly, these solutions y_1 and y_2 satisfy the differential equation because they are the functions that make the equation true. To make the simplest possible analogy, plugging y_1 or y_2 into $y'' + Ay' + By = 0$ will produce the true equation $0 = 0$, in the same way that plugging 1 into $x = 1$

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produces the true equation $1 = 1$. Something is a solution to an equation when plugging that something into the equation makes the equation true.

Let's do an example so that we can see these solution sets in action.

Example

Show that $\{y_1, y_2\} = \{e^{-3x}, xe^{-3x}\}$ are two solutions to the second order linear homogeneous equation.

$$y'' + 6y' + 9y = 0$$

To show that $y_1 = e^{-3x}$ is a solution to the differential equation, we'll need to start by taking the first and second derivatives of y_1.

$$y_1 = e^{-3x}$$

$$y_1' = -3e^{-3x}$$

$$y_1'' = 9e^{-3x}$$

Then we can plug these values into the homogeneous equation

$$y'' + 6y' + 9y = 0$$

$$9e^{-3x} + 6(-3e^{-3x}) + 9e^{-3x} = 0$$

$$9e^{-3x} - 18e^{-3x} + 9e^{-3x} = 0$$

$$(9 - 18 + 9)e^{-3x} = 0$$

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$$0e^{-3x} = 0$$

$$0 = 0$$

Because we end up with a true equation, we know that $y_1 = e^{-3x}$ satisfies the differential equation. Now let's check to make sure that $y_2 = xe^{-3x}$ does the same thing. Its first two derivatives are

$$y_2' = (1)(e^{-3x}) + (x)(-3e^{-3x})$$

$$y_2' = e^{-3x} - 3xe^{-3x}$$

and

$$y_2'' = -3e^{-3x} - ((3)(e^{-3x}) + (3x)(-3e^{-3x}))$$

$$y_2'' = -3e^{-3x} - (3e^{-3x} - 9xe^{-3x})$$

$$y_2'' = -3e^{-3x} - 3e^{-3x} + 9xe^{-3x}$$

$$y_2'' = -6e^{-3x} + 9xe^{-3x}$$

Then we can plug these values into the homogeneous equation

$$y'' + 6y' + 9y = 0$$

$$-6e^{-3x} + 9xe^{-3x} + 6(e^{-3x} - 3xe^{-3x}) + 9(xe^{-3x}) = 0$$

$$-6e^{-3x} + 9xe^{-3x} + 6e^{-3x} - 18xe^{-3x} + 9xe^{-3x} = 0$$

$$(-6 + 6)e^{-3x} + (9 - 18 + 9)xe^{-3x} = 0$$

$$0e^{-3x} + 0xe^{-3x} = 0$$

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$$0 = 0$$

This last example illustrates why $y_1 = e^{r_1 x}$ and $y_2 = xe^{r_2 x}$ are solutions to a differential equation $y'' + Ay' + By = 0$ whose characteristic equation has equal real roots.

In the same way, $\{y_1, y_2\} = \{e^{r_1 x}, e^{r_2 x}\}$ will be a solution set to a differential equation $y'' + Ay' + By = 0$ whose characteristic equation has distinct real roots. And $\{y_1, y_2\} = \{e^{\alpha x}\cos(\beta x), e^{\alpha x}\sin(\beta x)\}$ will be a solution set to a differential equation $y'' + Ay' + By = 0$ whose characteristic equation has complex conjugate roots.

Reduction of order with constant coefficients

Now that we better understand how y_1 and y_2 are solutions to the differential equation, we want to realize that we can use one solution to find the other.

In other words, if we already know y_1, we can use it to find y_2, or vice versa. This solution method is called **reduction of order**, because it uses a substitution to reduce the order of the homogeneous equation from a second order equation to a first order equation.

Depending on the kind of differential equation we're dealing with, it can be just as easy to find both solutions at once as it is to find only one solution. So we won't always use reduction of order to solve differential equations, but we still want to be familiar with this method.

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Let's do an example so that we can see how this works.

Example

Use reduction of order to find the general solution to the differential equation, given $y_1 = e^{x/3}$.

$$6y'' + y' - y = 0$$

To apply this method, we always begin with the assumption that $y_2 = vy_1$, and then we find the first and second derivatives of y_2, since we're dealing with a second order equation.

$$y_2 = ve^{\frac{x}{3}}$$

$$y_2' = v'e^{\frac{x}{3}} + \frac{1}{3}ve^{\frac{x}{3}}$$

$$y_2'' = v''e^{\frac{x}{3}} + \frac{2}{3}v'e^{\frac{x}{3}} + \frac{1}{9}ve^{\frac{x}{3}}$$

Plug these values into the homogeneous differential equation.

$$6v''e^{\frac{x}{3}} + 4v'e^{\frac{x}{3}} + \frac{2}{3}ve^{\frac{x}{3}} + v'e^{\frac{x}{3}} + \frac{1}{3}ve^{\frac{x}{3}} - ve^{\frac{x}{3}} = 0$$

$$6v''e^{\frac{x}{3}} + 5v'e^{\frac{x}{3}} = 0$$

This is when the reduction of order part comes in. We'll make a substitution $w = v'$, and therefore $w' = v''$.

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$$6w'e^{\frac{x}{3}} + 5we^{\frac{x}{3}} = 0$$

This substitution changes the second order equation in v into a first order equation in w. Furthermore, this first order equation is a linear equation that we can easily solve.

$$6e^{\frac{x}{3}}\frac{dw}{dx} + 5e^{\frac{x}{3}}w = 0$$

$$\frac{dw}{dx} + \frac{5e^{\frac{x}{3}}}{6e^{\frac{x}{3}}}w = 0$$

$$\frac{dw}{dx} + \frac{5}{6}w = 0$$

With $P(x) = 5/6$, the integrating factor is

$$I(x) = e^{\int P(x)\,dx}$$

$$I(x) = e^{\int \frac{5}{6}\,dx}$$

$$I(x) = e^{\frac{5}{6}x}$$

Multiplying the first order differential equation in w by the integrating factor, and then simplifying, gives

$$\frac{d}{dx}(we^{\frac{5}{6}x}) = 0$$

Integrate both sides.

$$\int \frac{d}{dx}(we^{\frac{5}{6}x})\,dx = \int 0\,dx$$

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$$we^{\frac{5}{6}x} = C$$

$$w = Ce^{-\frac{5}{6}x}$$

Because $w = v'$, we'll integrate w to find v.

$$v = \int w\, dx = \int Ce^{-\frac{5}{6}x}\, dx = -\frac{6}{5}Ce^{-\frac{5}{6}x} + k$$

We can choose any constants, so we'll choose $C = -5/6$ and $k = 0$ for simplicity. Then v is

$$v = e^{-\frac{5}{6}x}$$

and the second solution for the differential equation is

$$y_2 = ve^{\frac{1}{3}x}$$

$$y_2 = e^{-\frac{5}{6}x}e^{\frac{1}{3}x}$$

$$y_2 = e^{-\frac{x}{2}}$$

Because the solutions are $y_1 = e^{x/3}$ and $y_2 = e^{-x/2}$, the general solution to the differential equation is

$$y(x) = c_1e^{\frac{x}{3}} + c_2e^{-\frac{x}{2}}$$

Reduction of order with non-constant coefficients

In the previous example with $6y'' + y' - y = 0$, we applied reduction of order to a differential equation with constant coefficients. We say that the

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differential equation has constant coefficients because the the coefficients on y, y', and y'' are -1, 1, and 6, respectively. These are all constants.

But of course, we've seen previously that not all differential equations have constant coefficients, and reduction of order can be effective on these non-constant coefficient equations as well.

Specifically, reduction of order can help us find the second solution to an equation in the form

$$y'' + P(x)y' + Q(x)y = 0$$

when we already know one solution, y_1. Let's do an example so that we can see how this works.

Example

Use reduction of order to find the general solution to the differential equation, given $y_1 = x$.

$$x^2y'' - 4xy' + 4y = 0$$

To apply this method, we always begin with the assumption that $y_2 = vy_1$, and then we find the first and second derivatives of y_2, since we're dealing with a second order equation.

$$y_2 = vx$$

$$y_2' = v'x + v$$

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$$y_2'' = v''x + 2v'$$

Plug these values into the homogeneous differential equation.

$$x^2(v''x + 2v') - 4x(v'x + v) + 4vx = 0$$

$$v''x^3 + 2v'x^2 - 4v'x^2 - 4vx + 4vx = 0$$

$$v''x^3 - 2x^2v' = 0$$

This is when the reduction of order part comes in. We'll make a substitution $w = v'$, and therefore $w' = v''$.

$$w'x^3 - 2x^2w = 0$$

This substitution changes the second order equation in v into a first order equation in w. Furthermore, this first order equation is a linear equation that we can easily solve.

$$x^3\frac{dw}{dx} - 2x^2w = 0$$

$$\frac{dw}{dx} - \frac{2}{x}w = 0$$

With $P(x) = -2/x$ and $Q(x) = 0$, the integrating factor is

$$I(x) = e^{\int P(x)\,dx}$$

$$I(x) = e^{\int -\frac{2}{x}\,dx}$$

$$I(x) = e^{-2\ln x}$$

$$I(x) = x^{-2}$$

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Multiplying the first order differential equation in w by the integrating factor, and then simplifying, gives

$$\frac{d}{dx}(wx^{-2}) = 0$$

Integrate both sides.

$$\int \frac{d}{dx}(wx^{-2})\, dx = \int 0\, dx$$

$$wx^{-2} = C$$

$$w = Cx^2$$

Because $w = v'$, we'll integrate w to find v.

$$v = \int w\, dx = \int Cx^2\, dx = \frac{C}{3}x^3 + k$$

We can choose any constants, so we'll choose $C = 3$ and $k = 0$ for simplicity. Then v is

$$v = x^3$$

and the second solution for the differential equation is

$$y_2 = vx$$

$$y_2 = (x^3)x$$

$$y_2 = x^4$$

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Because the solutions are $y_1 = x$ and $y_2 = x^4$, the general solution to the differential equation is

$$y(x) = c_1 x + c_2 x^4$$

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